

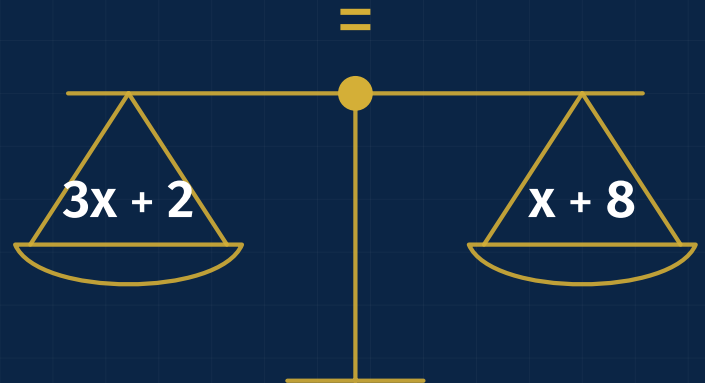


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CHAPTER 2 • ALGEBRA • TOPIC 10

Solving Quadratics

Calculator first — the answers tell you the method. Clean: factorise. Messy: the formula. Surd: complete the square.



7

VARIANTS

3

PAST-PAPER QUESTIONS

43

PRACTICE QUESTIONS

YOU WILL MASTER

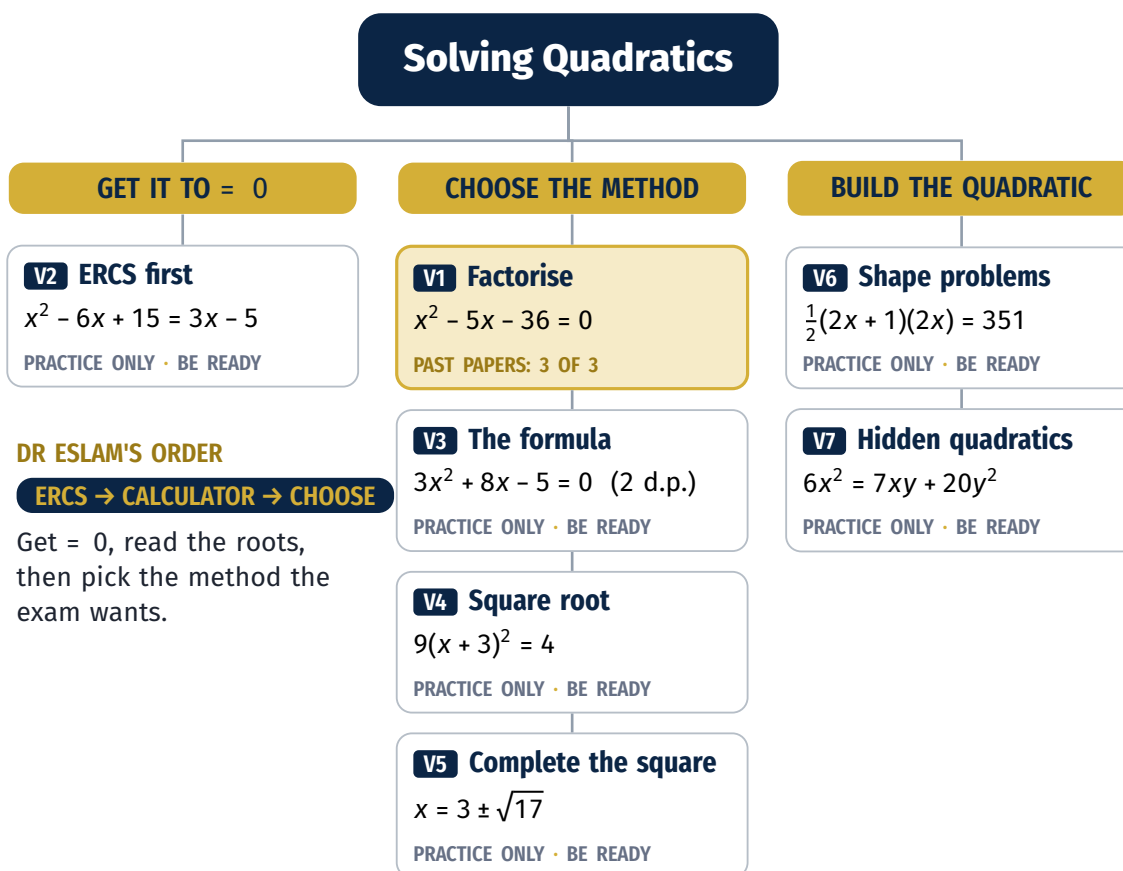
V1 Factorise V2 ERCS first V3 The formula V4 Square root

V5 Complete the square V6 Shape problems V7 Hidden quadratics

TOPIC MAP • EVERY VARIANT ON ONE PAGE

Solving Quadratics at a glance



Your notes


★ How often does each variant come up?

3 past-paper questions of their own (plus 14 “factorise, hence solve” parts in Topic 4); 43 practice parts — grouped by type.

	PAST PAPERS (OUT OF 3)		PRACTICE PARTS
V1 Factorise	● ● ●	3	9
V2 ERCS first	● ● ●	0	7
V3 The formula	● ● ●	0	11
V4 Square root	● ● ●	0	4
V5 Complete the square	● ● ●	0	4
V6 Shape problems	● ● ●	0	4
V7 Hidden quadratics	● ● ●	0	4

WHAT THIS TELLS YOU On the real papers, quadratics are solved by **factorising** (17 parts with Topic 4). The formula and surd form appear inside longer questions — be ready.

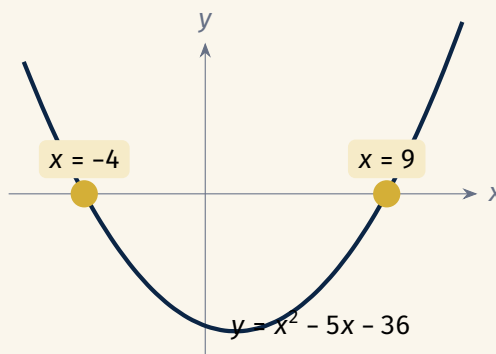
BEFORE YOU START

The big idea

**A quadratic has up to two answers.**

The highest power is x^2 . Its graph is a U-curve; the answers of $= 0$ are where the curve **crosses the x-axis**: usually **two**, sometimes one, sometimes none.

First make it $= 0$, then read the roots on the calculator.


Calculator first — the roots choose the method
Whole / simple fractions

$$x = 9, x = -4$$

Factorise (V1).
Brackets are the method mark.

Long decimals

$$x = 0.5197 \dots$$

The formula (V3).
Round at the end.

“Surd form”

$$x = 3 \pm \sqrt{17}$$

Complete the square (V5).

No x term

$$3x^2 = 75$$

Square root (V4),
 $\pm$.

ERCS — Dr Eslam’s first move

E EXPAND
every bracket

R REARRANGE
everything
to one side

C COLLECT
 $ax^2 + bx + c = 0$

S SOLVE
the right method

**CALCULATOR FIRST, THEN CHECK**

MENU → Equation → Polynomial → **2** (fx-991ES: **MODE** **5** **3**). Enter a, b, c — read x_1 and x_2 .

The calculator answer is your **check**, not your working: “show clear algebraic working” needs brackets or the formula line.

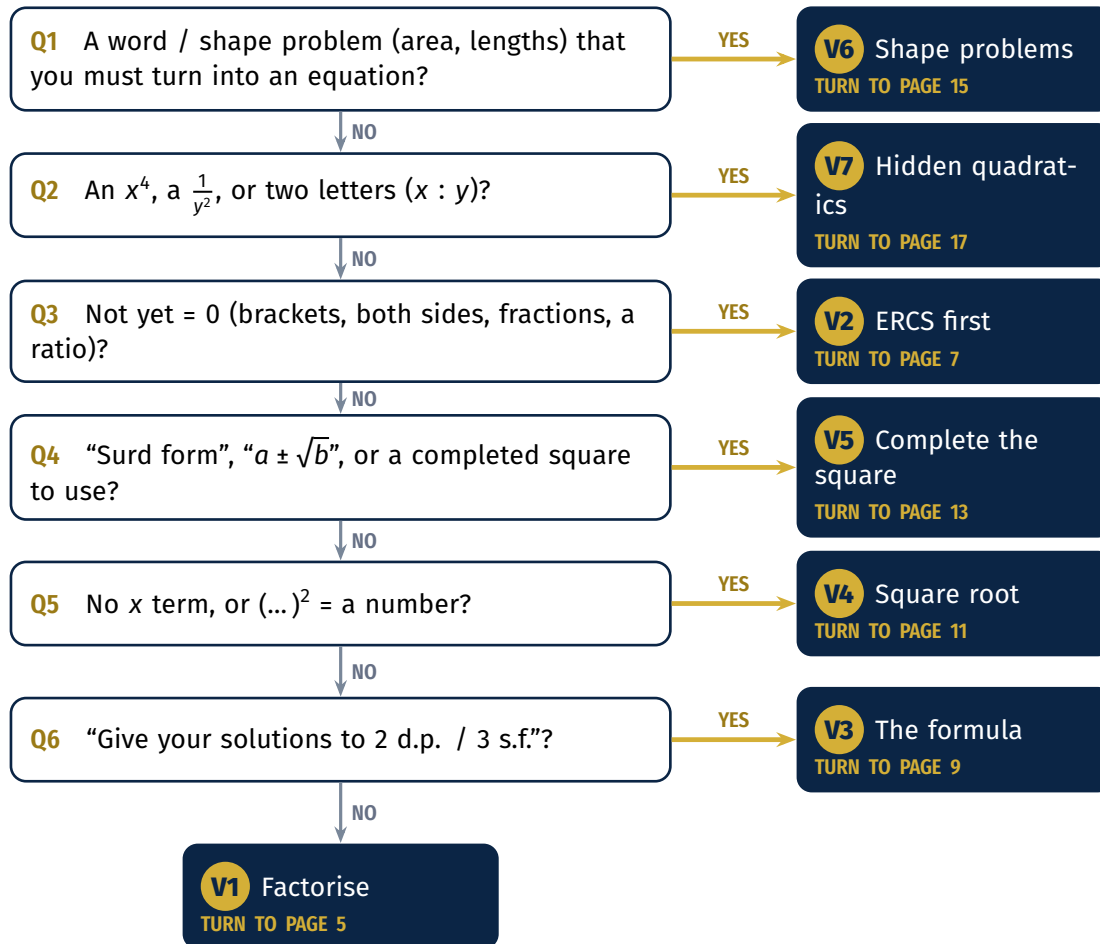
COUNT / RECOGNISE, THEN CHOOSE

Which variant am I?



Your notes

Read the question, then walk down the ladder. The first **YES** tells you the variant – and the page to turn to.

**COLOUR CODE IN THE WORKING**

Blue = letter terms $4x$, $-3y$

Orange = plain numbers 17 , -13

STEP 1 **STEP 2** **STEP 3** **STEP 4** = which step of “How do I do it?” each line is.

ON EVERY VARIANT SPREAD

- | | | | |
|---|-----------------|---|-----------------------|
|  | What is it? |  | When you see → you do |
|  | How do I do it? |  | Watch out |
|  | Worked example |  | Exam tip |
|  | Check yourself |  | Calculator check |

V1**VARIANT 1** MOST COMMON IN THE EXAM

Solve by Factorising

= 0 with clean roots: brackets, then each bracket = 0.

SEEN IN PAST PAPERS

●●● **3 / 3**

PRACTICE QUESTIONS: 9



Your notes

Came up in: Jan 2020 P1HR Q6 • Jan 2020 P2H Q7(b) • Jan 2021 P2HR Q9

? What is it?

In plain words

If the calculator roots are **whole numbers or simple fractions**, the quadratic factorises. Write the brackets (CC Rule: **flip the signs**), then **each bracket = 0**.

A root $-\frac{2}{5}$ gives the bracket $(5x + 2)$.

IT LOOKS LIKE

$$x^2 - 5x - 36 = 0$$

$$x^2 - 21x + 20 = 0$$

$$5x^2 + 7x + 2 = 0$$

$$x_1 = 9$$

$$x_2 = -4$$

↓ FLIP THE SIGNS

$$(x - 9)(x + 4) = 0$$

↓ EACH BRACKET = 0

$$x = 9 \text{ or } x = -4$$

How do I do it?

1 Calculator

Equation mode: read the roots. Clean? → factorise.

2 Brackets

Flip the signs: roots 9, -4 → $(x - 9)(x + 4)$. Fraction root $\frac{a}{b}$ → $(bx - a)$.

3 Each bracket = 0

Write $x - 9 = 0$ or $x + 4 = 0$.

4 Answers

$x = 9$ or $x = -4$ — they match the calculator.

CC RULE → EACH BRACKET = 0 Same as Topic 4, V5.

V1 Solve by Factorising

VARIANT 1 • SEE IT DONE, THEN TRY


 Your notes
WHEN YOU SEE → YOU DO

“Solve ... = 0. Show clear algebraic working.”

→ **Brackets line, then both answers**

“Solve by factorisation”

→ **Brackets are required — no formula**

“The only solution is $x = 5$ ”

→ **A double root:** $(x - 5)^2 = 0$

WORKED EXAMPLE **PAST PAPER**

January 2020 • Paper 1HR • Q6 • 3 marks

Solve $x^2 - 5x - 36 = 0$. Show clear algebraic working.

STEP 1 $x = 9, x = -4$

◀ calculator: clean roots

STEP 2 $(x - 9)(x + 4) = 0$

◀ brackets: flip the signs

STEP 3 $x - 9 = 0$ or $x + 4 = 0$

◀ each bracket = 0

STEP 4 $x = 9$ or $x = -4$

◀ solve

ANSWER $x = 9$ or $x = -4$

✓ $81 - 45 - 36 = 0$ ✓

WATCH OUT

✗ writing only “ $x = 9, -4$ ”

✓ $(x - 9)(x + 4) = 0$, so $x = 9$ or -4

“Show clear algebraic working” —
calculator answers alone score 0.

EXAM TIP

Three quick marks: M1 brackets, M1 each = 0, A1 both answers. Check each in the equation.

Check yourself

calculator first • answers on page 19

1.1 Jan 2020 P2H

Solve $x^2 - 3x - 40 = 0$

Answer:

1.2 Jan 2021 P2HR

Solve $x^2 - 21x + 20 = 0$

Answer:

1.3 Practice • Hard

Solve by factorisation
 $2x^2 - 19x - 33 = 0$

Answer:

V2**VARIANT 2**
ERCS First

The equation is not = 0 yet: expand,
rearrange, collect — then solve.

SEEN IN PAST PAPERS

● ● ● **0 / 3**

NOT ASKED YET • BE READY

PRACTICE QUESTIONS: 7


Your notes
? What is it?**In plain words**

Brackets, terms on both sides,
fractions, or a ratio — the calculator
cannot read it yet. Use **ERCS**:

Expand, **R**earrange (all to one side),
Collect to $ax^2 + bx + c = 0$, **S**olve.

A ratio $a : b = c : d$ becomes $ad = bc$.

IT LOOKS LIKE

$$x^2 - 6x + 15 = 3x - 5$$

$$x(3x - 9) = 4$$

$$x^2 : (3x + 5) = 1 : 2$$

$$x^2 - 6x + 15 = 3x - 5$$

↓ **R: $-3x + 5$ BOTH SIDES**

$$x^2 - 9x + 20 = 0$$

↓ **S: FACTORISE**

$$x = 4 \text{ or } x = 5$$

How do I do it?**1 Expand**

Every bracket. Fractions? $\times$ the bottoms. Ratio?
Cross-multiply.

2 Rearrange

Move every term to one side so the other side is 0.

3 Collect

Write $ax^2 + bx + c = 0$ — this line is often a mark by itself.

4 Solve

Calculator, then factorise (V1) or the formula (V3).

ERCS Expand • Rearrange • Collect • Solve.

V2 ERCS First

VARIANT 2 • SEE IT DONE, THEN TRY


 Your notes
 WHEN YOU SEE → YOU DO

A bracket: $x(3x - 9) = 4$ → **Expand, then -4 :** $3x^2 - 9x - 4 = 0$

Terms on both sides → **Everything to one side, $= 0$**

A ratio or fractions → **Cross-multiply / $\times$ the bottoms first**

 WORKED EXAMPLE **PRACTICE**

Solving Quadratics • Medium Q18 • 4 marks

Solve $x^2 - 6x + 15 = 3x - 5$

STEP 2 $x^2 - 6x + 15 - 3x + 5 = 0$ ◀ rearrange: all to the left

STEP 3 $x^2 - 9x + 20 = 0$ ◀ collect

STEP 4 $(x - 4)(x - 5) = 0$ ◀ solve: calculator 4, 5 → brackets

STEP 4 $x = 4$ or $x = 5$ ◀ each bracket = 0

ANSWER $x = 4$ or $x = 5$ ✓ $x = 4$: $16 - 24 + 15 = 7 = 12 - 5$ **WATCH OUT**

✗

$$x^2 - 6x + 15 = 3x - 5 \Rightarrow x^2 - 3x + 10 = 0$$

✓ $x^2 - 9x + 20 = 0$

Moving $3x$ makes $-3x$ (so $-9x$); moving -5 makes $+5$ (so $+20$).

 EXAM TIP

"Write in the form $ax^2 + bx + c = 0$ " is 1 mark on its own — always write the collected line.

 Check yourself

calculator first • answers on page 19

2.1 Dr EslamSolve $x(x + 3) = 10$

Answer:

2.2 Practice • Very hardSolve $x^2 = 4(x - 3)^2$

Answer:

2.3 Practice • Very hard

$x^2 : (3x + 5) = 1 : 2$.
Find the possible values of x .

Answer:

V3**VARIANT 3**

The Quadratic Formula

Messy roots or “to 2 d.p.”: substitute a, b, c into the formula.

SEEN IN PAST PAPERS

● ● ● **0 / 3**

NOT ASKED YET • BE READY

PRACTICE QUESTIONS: 11



Your notes

? What is it?

In plain words

If the calculator roots are long decimals, the quadratic **does not factorise**. Use the **formula** — it is in the formula booklet.

Write a, b, c first, put **negatives in brackets**, then the calculator.

IT LOOKS LIKE

$$3x^2 + 8x - 5 = 0 \text{ (2 d.p.)}$$

$$x^2 - 5x + 3 = 0 \text{ (3 s.f.)}$$

IN THE FORMULA BOOKLET

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

for $ax^2 + bx + c = 0$

How do I do it?

1 List

a, b, c from $ax^2 + bx + c = 0$ — signs included.

2 Substitute

Into the formula, negatives in brackets: $-(-4), (-4)^2$.

3 Calculator

Work out the + answer and the - answer separately.

4 Round

To the d.p. or s.f. asked. Two answers.

NEGATIVES IN BRACKETS The fraction line goes under everything.

V3 The Quadratic Formula

VARIANT 3 • SEE IT DONE, THEN TRY



Your notes
WHEN YOU SEE → YOU DO

“correct to 2 decimal places / 3 s.f.” → **The formula**

b is negative → **$-b$ is positive:** $-(-5) = 5$

“describe the error made” → **Check $-b$, $(b)^2$, and that $2a$ divides everything**

**WORKED EXAMPLE****PRACTICE**

Solving Quadratics • Medium Q17 • 4 marks

Solve $3x^2 + 8x - 5 = 0$. Give your solutions correct to 2 decimal places.

STEP 1

$$a = 3, \quad b = 8, \quad c = -5$$

◀ list

STEP 2

$$x = \frac{-8 \pm \sqrt{8^2 - 4(3)(-5)}}{2(3)} = \frac{-8 \pm \sqrt{124}}{6}$$

◀ substitute, brackets

STEP 3

$$x = 0.5225 \dots \quad \text{or} \quad x = -3.1892 \dots$$

◀ calculator: + and -

STEP 4

$$x = 0.52 \quad \text{or} \quad x = -3.19$$

◀ round to 2 d.p.

ANSWER $x = 0.52$ or $x = -3.19$

✓ equation mode gives the same

**WATCH OUT**

$$\times x = -8 \pm \frac{\sqrt{124}}{6}$$

$$\checkmark x = \frac{-8 \pm \sqrt{124}}{6}$$

The $2a$ divides the **whole** top, $-b$ included.

**EXAM TIP**

Write the substituted line — it is the method mark even if your rounding slips.

**Check yourself**

calculator first • answers on page 19

3.1

Practice • Medium

Solve $x^2 - 5x + 3 = 0$.
Give your answers to 3 s.f.

Answer:

3.2

Practice • Medium

Solve $3x^2 + 4x - 12 = 0$.
Give your answers to 2 d.p.

Answer:

3.3

Practice • Hard

Solve $5x^2 = 10x + 4$.
Give your answers to 2 d.p.

Answer:

V4

VARIANT 4

VARIANT 4

Square Root (no x term)

$3x^2 = 75$ or $9(x+3)^2 = 4$: get the square alone, then $\pm\sqrt{\quad}$.

SEEN IN PAST PAPERS

0 / 3

NOT ASKED YET • BE READY

PRACTICE QUESTIONS: 4



Your notes

? What is it?

In plain words

If there is **no x term** — or the x is inside one squared bracket — do not expand. Get the square alone, then take the square root: $\pm$, two answers.

Dr Eslam: **Move** → **Root** → **Solve**.

IT LOOKS LIKE

$$3x^2 = 75$$

$$(x - 2)^2 = 3$$

$$9(x + 3)^2 = 4$$

$$(x + 3)^2 = \frac{4}{9}$$

± ROOT

$$x + 3 = +\frac{2}{3}$$

$$x + 3 = -\frac{2}{3}$$

$$x = -\frac{7}{3}$$

$$X = -\frac{11}{3}$$

How do I do it?

1 Move

Get the square alone: divide by 9, or by 3.

2 Root

Square root both sides — **+ and -**.

3 Solve

Finish each branch: $x + 3 = \pm \frac{2}{3} \rightarrow$ two answers.

MOVE $\rightarrow$ ROOT ($\pm$) $\rightarrow$ SOLVE

“Without expanding” means exactly this.

V4 Square Root (no x term)

VARIANT 4 • SEE IT DONE, THEN TRY


 Your notes
WHEN YOU SEE → YOU DO

$3x^2 = 75$ (no x term)

$\rightarrow x^2 = 25, x = \pm 5$

"Without expanding ... exact solutions" $\rightarrow$ **Move $\rightarrow$ Root $\rightarrow$ Solve, keep fractions**

"Explain what is wrong with Liz's answer" ($x = 5$)

 $\rightarrow$ **She lost $x = -5$** **WORKED EXAMPLE****PRACTICE**

Solving Quadratics • Medium Q15 • 3 marks

Without expanding any brackets, work out the exact solutions of $9(x + 3)^2 = 4$.

STEP 1 $(x + 3)^2 = \frac{4}{9}$ $\leftarrow$ move: divide by 9

STEP 2 $x + 3 = \pm \frac{2}{3}$ $\leftarrow$ root: $\pm$

STEP 3 $x = -3 + \frac{2}{3} = -\frac{7}{3}$ or $x = -3 - \frac{2}{3} = -\frac{11}{3}$ $\leftarrow$ solve both

ANSWER $x = -\frac{7}{3}$ Or $x = -\frac{11}{3}$

$\checkmark 9\left(-\frac{2}{3}\right)^2 = 4 \checkmark$

**WATCH OUT**

$\times x^2 = 25 \Rightarrow x = 5$

$\checkmark x^2 = 25 \Rightarrow x = \pm 5$

$(-5)^2 = 25$ too. A square root always gives two answers.

**EXAM TIP**

"Exact" means keep fractions and surds — no decimals.

**Check yourself**

calculator first • answers on page 19

4.1

Practice • Medium

Solve $3x^2 = 75$

Answer:

4.2

Practice • Medium

Solve $(x - 2)^2 = 3$. Give your answers to 3 s.f.

Answer:

4.3

Practice • Medium

Liz solves $3x^2 + 8 = 83$:
 $3x^2 = 75, x^2 = 25, x = 5$.
 What is wrong?

Answer:

V5

VARIANT 5

Completing the Square to Solve

"Surd form" or " $a \pm \sqrt{b}$ ": complete the square, then Move → Root → Solve.

SEEN IN PAST PAPERS

● ● ● 0 / 3

NOT ASKED YET • BE READY

PRACTICE QUESTIONS: 4


Your notes

? What is it?

IN PLAIN WORDS

When the answer must be **exact with a surd** ($3 \pm \sqrt{17}$), complete the square first (Topic 5: list $a, b, c, \frac{b}{2a} \rightarrow$ structure). Then the letter appears once — finish with **Move → Root → Solve** as in V4.

IT LOOKS LIKE

$$x^2 - 6x - 8 = 0 \quad (a \pm \sqrt{b}) \quad x^2 + 4x - 16 = 0 \quad (\text{surd form})$$

COMPLETE THE SQUARE

MOVE

ROOT $\pm$

SOLVE

$$(x - 3)^2 - 17 = 0$$

$$(x - 3)^2 = 17$$

$$x - 3 = \pm\sqrt{17}$$

$$x = 3 \pm \sqrt{17}$$

Two moves from Topic 5 and V4 — the surd stays exact.

How do I do it?

1 Complete the square

$$x^2 - 6x - 8 = (x - 3)^2 - 9 - 8 = (x - 3)^2 - 17.$$

2 Move

$$(x - 3)^2 = 17.$$

3 Root

$$x - 3 = \pm\sqrt{17} \text{ — leave the surd.}$$

4 Solve

$$x = 3 \pm \sqrt{17}. \text{ Simplify surds: } \sqrt{20} = 2\sqrt{5}.$$

EXACT = KEEP THE SURD No decimals in surd form.

V5 Completing the Square to Solve

VARIANT 5 • SEE IT DONE, THEN TRY


 Your notes
 WHEN YOU SEE → YOU DO"in the form $a \pm \sqrt{b}$ " / "surd form"

→ Complete the square, then Move → Root → Solve

"Using your answer to part (a)"

→ Use the completed square — no formula

 $\sqrt{20}$, $\sqrt{108}$ in the answer→ Simplify: $2\sqrt{5}$, $6\sqrt{3}$ **WORKED EXAMPLE****PRACTICE**

Solving Quadratics • Hard Q1 • 3 marks

Solve $x^2 - 6x - 8 = 0$. Write your answer in the form $a \pm \sqrt{b}$ where a and b are integers.

STEP 1 $(x - 3)^2 - 9 - 8 = 0 \Rightarrow (x - 3)^2 - 17 = 0$ ◀ complete the square

STEP 2 $(x - 3)^2 = 17$ ◀ move

STEP 3 $x - 3 = \pm\sqrt{17}$ ◀ root, $\pm$

STEP 4 $x = 3 \pm \sqrt{17}$ ◀ solve

ANSWER $x = 3 \pm \sqrt{17}$ ($a = 3$, $b = 17$) ✓ 7.12 and -1.12: calculator agrees

**WATCH OUT**✗ $x = 3 \pm 17$ ✓ $x = 3 \pm \sqrt{17}$

The square root of 17 is not 17 — keep the root sign.

**EXAM TIP**

Check your surd answer with the calculator's decimals: $3 + \sqrt{17} = 7.12 \dots$

**Check yourself**

calculator first • answers on page 19

5.1

Practice • Hard

$x^2 + 4x - 16 = (x+2)^2 - 20$.
Solve $x^2 + 4x - 16 = 0$ in surd form.

Answer:

5.2

Practice • Very hard

$7 - 4x - x^2 = 11 - (x+2)^2$.
Use it to solve
 $7 - 4(y+3) - (y+3)^2 = 0$.

Answer:

5.3

Dr Eslam

Solve $x^2 - 4x - 3 = 0$,
giving your answers in surd form.

Answer:

V6

VARIANT 6

Shape Problems

An area is given: build the equation from the shape, show it, solve, reject the negative.

SEEN IN PAST PAPERS

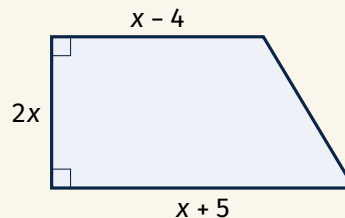
● ● ● 0 / 3

NOT ASKED YET • BE READY

PRACTICE QUESTIONS: 4


 Your notes
? What is it?**In plain words**

The sides of a shape are expressions in x . Use the **area formula** with the given area, expand and rearrange to the equation they want ("show that"). Solve it — a length cannot be negative, so **reject** the negative answer.



$$\begin{aligned}\text{area} &= \frac{1}{2}(a + b)h \\ &= \frac{1}{2}(2x + 1)(2x) = 351\end{aligned}$$

IT LOOKS LIKE

trapezium area 351

rectangle area 60

How do I do it?**1 Formula**

Write the area (or perimeter) formula for the shape.

2 Substitute & show

Put the expressions in, = the value; ERCS to the given equation.

3 Solve

Calculator → factorise or the formula.

4 Reject

A length must be positive: reject the negative root.

SHAPE → EQUATION → SOLVE → REJECT Check the sides are positive.

V6 Shape Problems

VARIANT 6 • SEE IT DONE, THEN TRY



Your notes
WHEN YOU SEE → YOU DO

"The area is 351 cm^2 . Show that ..."

→ **Area formula = 351, then ERCS**

"Work out the value of x " (a length)

→ **Reject the negative answer**

Two shapes with equal areas

→ **Set the two area expressions equal**

**WORKED EXAMPLE****PRACTICE**

Solving Quadratics • Very hard Q4 • 5 marks

A trapezium has parallel sides $(x - 4)$ and $(x + 5)$ and height $2x$. Its area is 351 cm^2 . (a) Show that $2x^2 + x - 351 = 0$. (b) Find x .

- STEP 1** $\frac{1}{2}((x - 4) + (x + 5))(2x) = 351$ ◀ area formula
- STEP 2** $x(2x + 1) = 351 \Rightarrow 2x^2 + x - 351 = 0$ ◀ (a) expand, rearrange
- STEP 3** $(2x + 27)(x - 13) = 0$ ◀ (b) calculator: 13, -13.5
- STEP 4** $x = 13$ ($x = -13.5$ rejected) ◀ a length is positive

ANSWER $x = 13$ ✓ sides 9, 18, height 26: $\frac{1}{2}(27)(26) = 351$ **WATCH OUT**✗ giving both $x = 13$ and $x = -13.5$ ✓ $x = 13$ only

A side of $-13.5 - 4 \text{ cm}$ is impossible — say you reject it.

**EXAM TIP**

In "show that", do not start from the answer — start from the area formula and end at the given equation.

**Check yourself**

calculator first • answers on page 19

6.1

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A rectangle is $x \text{ cm}$ by $(x + 4) \text{ cm}$, area 60 cm^2 . Find the width.

Answer:

6.2

Practice • Very hard

Trapezium: parallel sides $(x + 5)$, $(3x - 2)$, height $(2x - 3)$, area 133 cm^2 . Find x .

Answer:

6.3

Practice • Hard

A square has side $(3 + x) \text{ cm}$ and area 10 cm^2 . Show that $x^2 + 6x = 1$.

Answer:

V7**VARIANT 7****Hidden Quadratics**

x^4 , $\frac{1}{y^2}$, or two letters: a substitution or a factorisation reveals a quadratic.

SEEN IN PAST PAPERS

● ● ● **0 / 3**

NOT ASKED YET • BE READY

PRACTICE QUESTIONS: 4



Your notes
? What is it?**In plain words**

x^4 and x^2 : let $u = x^2 \rightarrow$

$$6u^2 + 7u - 10 = 0.$$

$\frac{1}{y^2}$ and $\frac{1}{y}$: let $u = \frac{1}{y}$.

Two letters ($6x^2 = 7xy + 20y^2$):

factorise with x and $y \rightarrow$ a ratio $x : y$.

IT LOOKS LIKE

$$6x^4 + 7x^2 = 10$$

$$\frac{2}{y^2} + \frac{9}{y} - 7 = 0$$

$$6x^2 = 7xy + 20y^2$$

$$6(x^2)^2 + 7x^2 = 10$$

↓ **LET $u = x^2$**

$$6u^2 + 7u - 10 = 0$$

solve for u , then $x^2 = u$

How do I do it?**1 Spot the pattern**

Powers x^4 and x^2 ? $\frac{1}{y^2}$ and $\frac{1}{y}$? Two letters?

2 Substitute / factorise

$u = x^2$ (or $u = \frac{1}{y}$); or factorise in x and y .

3 Solve the quadratic

Calculator $\rightarrow$ factorise or formula, for u .

4 Back to the letter

$x^2 = u$, $y = \frac{1}{u}$; reject impossible values ($x^2 < 0$).

LET $u = x^2$ Then go back to the real letter.

V7 Hidden Quadratics

VARIANT 7 • SEE IT DONE, THEN TRY

 Your notes

WHEN YOU SEE → YOU DO

 x^4 and x^2 only→ Let $u = x^2$ $"x > 0, y > 0$. Find the ratio $x : y"$ → Factorise: $(2x - 5y)(3x + 4y) = 0$ $\frac{2}{y^2} + \frac{9}{y} - 7 = 0$ → Let $u = \frac{1}{y}$ (or xy^2)

WORKED EXAMPLE PRACTICE

Solving Quadratics • Very hard Q12 • 3 marks

 $6x^2 = 7xy + 20y^2$ where $x > 0$ and $y > 0$. Find the ratio $x : y$.

STEP 1 $6x^2 - 7xy - 20y^2 = 0$

◀ rearrange

STEP 2 $(2x - 5y)(3x + 4y) = 0$

◀ factorise in x and y

STEP 3 $3x + 4y \neq 0$ ($x, y > 0$) $\Rightarrow 2x = 5y$

◀ reject the impossible bracket

STEP 4 $x : y = 5 : 2$

◀ ratio

ANSWER $x : y = 5 : 2$ ✓ $x = 5, y = 2: 150 = 70 + 80$

WATCH OUT

✗ $u = \frac{5}{6}$, so $x = \frac{5}{6}$

✓ $u = x^2 = \frac{5}{6}$

After substituting, go back: u is not the answer.

EXAM TIP

Try the calculator with $y = 1$:

$6x^2 - 7x - 20 = 0$ gives $x = 2.5$, so

$x : y = 2.5 : 1 = 5 : 2$.

Check yourself

calculator first • answers on page 19

7.1 Practice • Very hard

$y = 6x^4 + 7x^2$ and
 $x = \sqrt{w + 1}$. Find w when
 $y = 10$.

Answer:

7.2 Practice • Very hard

Solve $\frac{2}{y^2} + \frac{9}{y} - 7 = 0$ to 3
 s.f.

Answer:

7.3 Practice • Very hard

$(2x - 1) : (x - 4) =$
 $(16x + 1) : (2x - 1)$. Find
 the possible values of
 x .

Answer:

CHECK YOURSELF

Answers

Every answer below was checked with sympy (check_answers.py). Got one wrong? Go back to that variant's Watch-out box first.

$$1.1 \quad x = 8 \text{ or } x = -5$$

$$1.2 \quad x = 1 \text{ or } x = 20$$

$$1.3 \quad x = -\frac{3}{2} \text{ or } x = 11$$

$$2.1 \quad x = 2 \text{ or } x = -5$$

$$2.2 \quad x = 2 \text{ or } x = 6$$

$$2.3 \quad x = 2.5 \text{ or } x = -1$$

$$3.1 \quad x = 4.30 \text{ or } x = 0.697$$

$$3.2 \quad x = 1.44 \text{ or } x = -2.77$$

$$3.3 \quad x = 2.34 \text{ or } x = -0.34$$

$$4.1 \quad x = \pm 5$$

$$4.2 \quad x = 3.73 \text{ or } x = 0.268$$

$$4.3 \quad \text{She lost } x = -5: x = \pm 5$$

$$5.1 \quad x = -2 \pm 2\sqrt{5}$$

$$5.2 \quad y = -5 \pm \sqrt{11}$$

$$5.3 \quad x = 2 \pm \sqrt{7}$$

$$6.1 \quad \text{width } 6 \text{ cm}$$

$$6.2 \quad x = 6.25$$

$$6.3 \quad (3+x)^2 = 10 \Rightarrow x^2 + 6x = 1$$

$$7.1 \quad w = -\frac{1}{6}$$

$$7.2 \quad y = 1.48 \text{ or } y = -0.193$$

$$7.3 \quad x = 5 \text{ or } x = -\frac{1}{12}$$

THE WHOLE TOPIC ON ONE PAGE

Solving Quadratics cheat sheet

**V1 Factorise 3/3**

$$(x - 9)(x + 4) = 0 \Rightarrow x = 9, -4$$

Clean roots $\rightarrow$ brackets $\rightarrow$ each = 0.

V2 ERCS first

$$x^2 - 6x + 15 = 3x - 5 \rightarrow x^2 - 9x + 20 = 0$$

Expand • Rearrange • Collect • Solve.

V3 The formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Messy roots, "2 d.p.". Negatives in brackets.

V4 Square root

$$(x + 3)^2 = \frac{4}{9} \Rightarrow x + 3 = \pm \frac{2}{3}$$

No x term: Move $\rightarrow$ Root ($\pm$) $\rightarrow$ Solve.

V5 Complete the square

$$(x - 3)^2 = 17 \Rightarrow x = 3 \pm \sqrt{17}$$

"Surd form": keep the root.

V6 Shape problems

$$x(2x + 1) = 351$$

Area formula, ERCS, reject the negative.

V7 Hidden quadratics

$$u = x^2; (2x - 5y)(3x + 4y) = 0$$

Substitute or factorise, then go back.

**TOP 3 LOST MARKS**

1. calculator answers only $\times$

2. $x^2 = 25 \Rightarrow x = 5$ only $\times$

3. $-b \pm \frac{\sqrt{}}{2a} \times$

BOTTOM LINE ERCS $\rightarrow$ calculator roots $\rightarrow$ clean: factorise • messy: formula • surd: complete the square $\rightarrow$ two answers.